APO-INDAPAMIDE TABLETS
Indapamide USP 2.5mg Diuretic / Antihypertensive

PHARMACOLOGY: Indapamide is a diuretic antihypertensive agent. The mechanism whereby indapamide exerts its action in the control of hypertension is not completely elucidated: both renal and extra renal actions may be involved. The renal site of action is the proximal part of the distal tubule and the ascending part of Henle's loop. Sodium and chloride ions are excreted in approximately equivalent amounts. The increased delivery of sodium to the distal tubular exchange site results in increased potassium excretion and hypokalemia.

Indapamide is rapidly and completely absorbed after oral administration. Peak blood levels are obtained after 1 to 2 hours. Indapamide is concentrated in the erythrocytes and is 79% bound to plasma proteins and to erythrocytes.

It is taken up by the vascular wall in smooth vascular muscle according to its high lipid solubility. Seventy per cent of a single oral dose is eliminated by the kidneys and 23 per cent by the gastrointestinal tract. Indapamide is metabolized to a marked degree, the unchanged product representing approximately 5 per cent of the total dose found in the urine during the 48 hours following administration. Elimination of indapamide from the plasma is biphasic with half-lives of 14 and 25 hours respectively.

INDICATIONS: APO-INDAPAMIDE (indapamide) is indicated in the management of essential hypertension. It may be tried as a sole therapeutic agent in the treatment of mild to moderate hypertension. Normally APO-INDAPAMIDE, as other diuretics, is used as the initial agent in multiple drug regimens.

CONTRAINDICATIONS: Hypersensitivity to indapamide, to other sulfonamides or to any of the excipients, Severe renal failure (creatinine clearance below 30 ml/min), Hepatic encephalopathy or severe impairment of liver function, Hypokalaemia, Combinations with anti-arrhythmic agents causing torsade de pointes, Pregnancy, Lactation, Hereditary problems of galactose intolerance, glucose-galactose malabsorption, or total lactase deficiency as APO–INDAPAMIDE contains lactose.

WARNINGS AND PRECAUTIONS:
<u>Electrolyte changes and hypokalaemia</u>
- Electrolyte changes observed with indapamide may be severe. The recommended maximum daily dose of 2.5mg/day should not be exceeded.
- Hypokalaemia may occur at all doses with consequent weakness, cramps and cardiac dysrhythmias. Hypokalaemia is a particular hazard in digitalized patients; dangerous or fatal cardiac arrhythmias may be precipitated.
- Hypokalaemia occurs commonly with diuretics; electrolyte monitoring is essential particularly in patients who would be at increased risk from hypokalaemia, such as patients with cardiac arrhythmias or those who are receiving concomitant cardiac glycosides.
- Potassium depletion with hypokalaemia is the major risk of thiazide and related diuretics. Hypokalaemia may cause muscle disorders. Cases of Rhabdomyolysis have been reported, mainly in the context of severe hypokalaemia. The risk of onset of hypokalaemia (< 3.4 mmol/l) must be prevented in certain high risk populations, i.e. the elderly, malnourished and/or polymedicated, cirrhotic patients with oedema and ascites, coronary artery disease and cardiac failure patients. In this situation, hypokalaemia increases the cardiac toxicity of digitalis preparations and the risks of arrhythmias.

- Individuals with a long QT interval are also at risk, whether the origin is congenital or iatrogenic. Hypokalaemia and bradycardia are both predisposing factors to the onset of severe arrhythmias, in particular, potentially fatal *torsades de pointes* among those individuals with long QT interval.
- More frequent monitoring of plasma potassium is required in all the situations indicated above. The first measurement of plasma potassium should be obtained during the first week following the start of treatment. Detection of hypokalaemia requires its correction.
- Patients receiving APO–INDAPAMIDE (indapamide) should be carefully observed and serum electrolytes monitored for signs and symptoms of fluid or electrolyte imbalance; namely hyponatremia, hypochloremia and hypokalaemia. Blood urea nitrogen, uric acid, and glucose levels should also be assessed during therapy. Hypokalaemia, an ever present hazard with most diuretics, will be more common in association with concomitant steroid or ACTH therapy and with inadequate electrolyte intake. The serum potassium should be determined at regular intervals and potassium supplementation instituted when indicated. (See Warnings).
- The signs of electrolyte imbalance are: dryness of the mouth, thirst, weakness, lethargy, drowsiness, restlessness, muscle pains or cramps, muscle fatigue, hypotension, oliguria, gastrointestinal disturbances such as nausea and vomiting, tachycardia and ECG changes.

<u>Monitoring of sodium and other electrolytes</u>

- Plasma sodium must be measured before starting treatment, then at regular intervals subsequently. Any diuretic treatment may cause hyponatraemia, sometimes with very serious consequences. The fall in plasma sodium may be asymptomatic initially and regular monitoring is therefore essential, and should be even more frequent in the elderly and cirrhotic patients.
- Calcium excretion is decreased by diuretics pharmacologically related to indapamide. However, after six to eight weeks of indapamide 1.25 mg treatment and in long-term studies of hypertensive patients with higher doses of indapamide, serum concentrations of calcium increased only slightly with indapamide.
- Prolonged treatment with drugs pharmacologically related to indapamide may in rare instances be associated with hypercalcemia and hypophosphatemia secondary to physiologic changes in the parathyroid gland; however, the common complications of hyperparathyroidism, such as renal lithiasis, bone resorption, and peptic ulcer, have not been seen. Frank hypercalcaemia may be due to previously unrecognised hyperparathyroidism.
- Treatment should be discontinued before tests for parathyroid function are performed. Like the thiazides, indapamide may decrease serum PBI levels without signs of thyroid disturbance.

<u>Renal Impairment and Function</u>

- Patients with renal insufficiency receiving APO-INDAPAMIDE (indapamide) should be carefully monitored. If an increase of azotemia and oliguria occur during treatment, the diuretic should be discontinued.
- Thiazide and related diuretics are fully effective only when renal function is normal or only minimally impaired (plasma creatinine below levels of the order of 25 mg/l, *i.e*. 220 mcmol/l in an adult) but are ineffective when the creatinine clearance is less than 30 mL/min.
- In the elderly, this plasma creatinine must be adjusted in relation to age, weight and gender. Hypovolaemia, secondary to the loss of water and sodium induced by the diuretic at the start of treatment causes a reduction in glomerular filtration. This may lead to an increase in blood urea and plasma creatinine. This transitory functional renal insufficiency is of no consequence in individuals with normal renal function but may worsen preexisting renal insufficiency.

<u>Hepatic Impairment</u>

- Indapamide is contraindicated in patients with severe hepatic impairment since diuretics may induce metabolic alkalosis in cases of potassium depletion which may precipitate episodes of hepatic encephalopathy (See Contraindications).
- When liver function is impaired, thiazide-related diuretics may cause, particularly in case of electrolyte imbalance, hepatic encephalopathy which can progress to hepatic coma. Administration of the diuretic should be stopped immediately if this occurs.

Hyperuricemia and Gout
- Hyperuricemia may occur during administration of APO-INDAPAMIDE. Rarely gout has been reported. Blood uric acid levels should be monitored, particularly in patients with a history of gout, who should continue to receive appropriate treatment.

Photosensitivity
- Cases of photosensitivity reactions have been reported with thiazides and thiazide-related diuretics (see Adverse Reactions). If photosensitivity reaction occurs during treatment, it is recommended to stop the treatment. If a re-administration of the diuretic is deemed necessary, it is recommended to protect exposed areas to the sun or to artificial UVA.

Blood Pressure and Hypotension Effects
- Orthostatic hypotension may occur and may be potentiated by alcohol, barbiturates, narcotics or concurrent therapy with other antihypertensives.
- The antihypertensive effect of the drug may be enhanced in the patient postsympathectomy.
- When indapamide is given with other non – diuretic antihypertensive agents, the effects on blood pressure are additive.

Other Metabolic and Systemic Effects
- Although indapamide exerts minimal effect on glucose metabolism, insulin requirements may be affected in diabetics. Monitoring of blood glucose is important in diabetics, in particular in the presence of hypokalaemia. Hyperglycemia and glycosuria may occur in patients with latent diabetes.
- Sulfonamide derivatives have been reported to exacerbate or activate systemic lupus erythematosus. These possibilities should be kept in mind with the use of indapamide although no case has been reported to date.

Dermatological Reactions: Severe dermatological adverse reactions, some accompanied by systemic manifestations, have been rarely reported with the use of indapamide. In the majority of cases, the condition subsided within 14 days following discontinuation of indapamide therapy (See Adverse Reactions).

Choroidal effusion, acute myopia and secondary angle-closure glaucoma: Sulfonamide or sulfonamide derivative drugs can cause an idiosyncratic reaction resulting in choroidal effusion with visual field defect, transient myopia and acute angle-closure glaucoma. Symptoms include acute onset of decreased visual acuity or ocular pain and typically occur within hours to weeks of drug initiation. Untreated acute angle-closure glaucoma can lead to permanent vision loss. The primary treatment is to discontinue drug intake as rapidly as possible. Prompt medical or surgical treatments may need to be considered if the intraocular pressure remains uncontrolled. Risk factors for developing acute angle-closure glaucoma may include a history of sulfonamide or penicillin allergy.

Use in Pregnancy: As a general rule, the administration of diuretics should not be used by pregnant women nor used to treat physiological oedema of pregnancy. Diuretics can cause foetoplacental ischaemia, with a risk of impaired foetal growth
Use in Nursing Mothers: Breast-feeding is inadvisable (Indapamide is excreted in human milk).
Use in Children: The safety and effectiveness of indapamide in children have not been established. Its use in this age group, therefore, is not recommended.
Use in Elderly: In the elderly, the plasma creatinine must be adjusted in relation to age, weight and gender. Elderly patients can be treated with indapamide 1.25 mg when renal function is normal or only minimally impaired.
Use in Athletes: Athletes should note that this product contains an active substance which may cause a positive reaction in doping tests.

Indapamide does not affect vigilance but different reactions in relation with the decrease in blood pressure may occur in individual cases, especially at the start of the treatment or when another antihypertensive agent is added. As a result the ability to drive vehicles or to operate machinery may be impaired.

DRUG INTERACTIONS:
Combinations that are not recommended:
- Lithium: Increased plasma lithium with signs of overdosage, as with a salt-free diet (decreased urinary lithium excretion). However, if the use of diuretics is necessary, careful monitoring of plasma lithium and dose adjustment are required.

Combinations requiring precautions for use:
- Torsades de pointes-inducing drugs such as but not limited to:
    - class Ia antiarrhythmic agents (e.g. quinidine, hydroquinidine, disopyramide) and class Ic (e.g. flecainide),
    - class III antiarrhythmic agents (e.g. amiodarone, sotalol, dofetilide, ibutilide),
    - some antipsychotics: phenothiazines (e.g. chlorpromazine, levomepromazine, trifluoperazine), benzamides (e.g. amisulpride), butyrophenones (e.g. haloperidol), other antipsychotics (e.g. pimozide)
    - psychoanaleptic (e.g. donepezil),
    - SSRIs (e.g. citalopram, escitalopram),
    - antimicrobial agents: fluoroquinolones (e.g. moxifloxacin, ciprofloxacin), macrolides, (e.g. erythromycin IV, clarithromycin), azole antifungals (e.g. fluconazole),
    - antiparasitics (e.g. chloroquine, pentamidine)
    - antihistamines
    - antiemetics (e.g. ondansetron, domperidone)
    - antineoplastic and immunomodulating agents (e.g. vandetanib, oxaliplatin, anagrelide),
    - anaesthetics (e.g. propofol, sevoflurane)
    - other substances such as: bepridil, methadone, papaverine.

  Increased risk of ventricular arrhythmias, particularly torsades de pointes (hypokalaemia is a risk factor). Monitor for hypokalaemia and correct, if required, before introducing this combination. Clinical, plasma electrolytes and ECG monitoring. Use substances which do not have the disadvantage of causing torsades de pointes in the presence of hypokalaemia.
- NSAID. (systemic route) including COX-2 selective inhibitors, high dose salicylic acid (≥3 g/day): Possible reduction in the antihypertensive effect of indapamide. Risk of acute renal failure in dehydrated patients (decreased glomerular filtration). Hydrate the patient; monitor renal function at the start of

treatment.

- Antihypertensive medications (e.g. Angiotensin converting enzyme (ACE) inhibitors): Risk of sudden hypotension and/or acute renal failure when treatment with an ACE inhibitor is initiated in the presence of preexisting sodium depletion (particularly in patients with renal artery stenosis). In hypertension, when prior diuretic treatment may have caused sodium depletion, it is necessary either to stop the diuretic 3 days before starting treatment with the ACE inhibitor, and restart a hypokalaemic diuretic if necessary or give low initial doses of the ACE. inhibitor and increase the dose gradually. In congestive heart failure, start with a very low dose of ACE inhibitor, possibly after a reduction in the dose of the concomitant hypokalaemic diuretic. In all cases, monitor renal function (plasma creatinine) during the first weeks of treatment with an ACE inhibitor.
- Other compounds causing hypokalaemia (e.g. amphotericin B (IV), corticosteroids (e.g. gluco and mineralo corticoids), ACTH (e.g. tetracosactide) and stimulant laxatives: Increased risk of hypokalaemia (additive effect). Monitoring of plasma potassium and correction if required. Use non-stimulant laxatives.
- Skeletal muscle relaxants (e.g. Baclofen): Increased antihypertensive effect. Hydrate the patient; monitor renal function at the start of treatment.
- Digitalis preparations (e.g. Digoxin): Hypokalaemia predisposing to the toxic effects of digitalis. Monitoring of plasma potassium and ECG and, if necessary, adjust the treatment.
- Citalopram: Increased risk of hyponatraemia.
- Allopurinol: Concomitant treatment with indapamide may increase the incidence of hypersensitivity reactions to allopurinol.

Combinations to be taken into consideration:

- Potassium-sparing diuretics (amiloride, spironolactone, triamterene): Whilst rational combinations are useful in some patients, hypokalaemia or hyperkalaemia (particularly in patients with renal failure or diabetes) may still occur. Plasma potassium and ECG should be monitored and, if necessary, treatment reviewed.
- Antihyperglycemic medications (e.g. Metformin): Increased risk of metformin induced lactic acidosis due to the possibility of functional renal failure associated with diuretics and more particularly with loop diuretics. Do not use metformin when plasma creatinine exceeds 15 mg/l (135 μ mol/l) in men and 12 mg/l (110 μ mol/l) in women.
- Iodinated contrast media: In the presence of dehydration caused by diuretics, increased risk of acute renal failure, in particular when large doses of iodinated contrast media are used. Rehydration before administration of the iodinated compound.
- Anticholinergic and adrenergic antagonists (e.g. Imipramine-like antidepressants, neuroleptics in particular pimozide): Antihypertensive effect and risk of orthostatic hypotension increased (additive effect).
- Calcium (salts): Risk of hypercalcaemia resulting from decreased urinary elimination of calcium.
- Ciclosporin, tacrolimus: Risk of increased plasma creatinine without any change in circulating cyclosporine levels, even in the absence of water/sodium depletion.
- Corticosteroids: Decreased antihypertensive effect (water/sodium retention due to corticosteroids).

ADVERSE REACTIONS:

The most severe and common adverse effect is electrolyte imbalance. Electrolyte changes reported include:

- Hypokalemia - 14.2% (requiring K+ supplementation 6%; with clinical symptoms 1.2%)
- Hypochloremia - 9.4%
- Hyponatremia - 3.1%

The other changes observed in laboratory parameters are minor and infrequent: elevation in blood uric acid (8.6%), blood glucose (6.0%), BUN (5.7%), and blood creatinine (3.6%).

The most frequent adverse events (incidence ≥ 1%) reported for patients treated with indapamide 2.5 mg were: headache (3.4%), vertigo (2.2%), dizziness (1.9%), asthenia (1.7%) and muscle cramps (1.2%).
In another study, the most frequently reported adverse events (incidence ≥ 1%) in the indapamide 1.25 mg group were: headache (17%), infection (12%), pain (8%), dizziness (7%), back pain (5%), rhinitis (5%), asthenia (4%), dyspepsia (4%), flu syndrome (3%), hypertonia (3%), sinusitis (3%), chest pain (2%), constipation (2%), cough (2%), diarrhoea (2%), edema (2%), nausea (2%), pharyngitis (2%), conjunctivitis (1%), nervousness (1%) and ECG abnormalities (non-specific ST-T changes (7%), sinus bradycardia (3%), arrhythmia (2%) or tachycardia (2%)).

All other adverse events occured at an incidence of less than 1% and included by body system:
Central Nervous: Drowsiness, sleepiness, insomnia, weakness, lethargy, visual disturbance, agitation, amnesia, anxiety, ataxia, coordination abnormality, depression, dream abnormality, hyperesthesia, insomnia, migraine, paresthesia, somnolence, twitching and vertigo.
Gastrointestinal: Nausea, anorexia, dryness of mouth, gastralgia, vomiting, diarrhea, constipation, increased appetite, GI carcinoma, GI disorders, duodenitis, dysphagia, esophagitis, flatulence, gastritis, gastroenteritis, oral moniliasis, proctitis, rectal disorders, rectal hemorroids, stomatitis, tooth disorders.
Musculoskeletal: Joint pain, back pain, weakness of legs, arthralgia, arthritis, bone disorders, joint disorders, bone fracture, bone pain, chondrodystrophy, myalgia, myasthenia and myopathy.
Cardiovascular: Orthostatic hypotension, tachycardia, ECG changes (non-specific ST-T change, U waves, left ventricular strain), angina pectoris, bundle branch block, ventricular extrasystoles, atrial fibrillation, atrial flutter, hypertension, postural hypotension, palpitations, syncope, supraventricular tachycardia and vasodilation.
Urogenital: Impotence, modification of libido, polyuria, dysmenorrhea, dysuria, impotence, urinary tract infection, nocturia, oliguria, urinary frequency or urgency, renal pain or calculus, prostate disorders and vaginitis.
Respiratory: bronchitis, dyspnea, laryngitis, lung disorder and sputum increase.
Dermatological: Rash, pruritus, acne, application site reaction, exfoliative dermatitis, nail disorder, skin nodule, rash, bullous eruption and sweat.
Endocrine: Gout, diabetes mellitus
Special senses: amblyopia, ear disorders, ear pain, otitis, photophobia, taste perversion, tinnitus and vision abnormality.
Other: Tinnitus, malaise / fainting, sweat, thyroid disorder, ecchymosis, allergic reaction, edema face, fever, hernia, malaise and monilia.

Postmarketing experience:
The most commonly reported adverse reactions are hypersensitivity reactions, mainly dermatological, in subjects with a predisposition to allergic and asthmatic reactions and maculopapular rashes.
Nervous system disorders: fatigue
Cardiac disorders: electrocardiogram QT prolonged, torsade de pointes (potentially fatal)
Hepato-biliary disorders: hepatitis
Investigations: elevated liver enzyme levels
Metabolism and nutrition disorders: Hypercalcaemia, Hyponatraemia with hypovolaemia responsible for dehydration and orthostatic hypotension. Concomitant loss of chloride ions may lead to secondary compensatory metabolic alkalosis: the incidence and degree of this effect are slight.

Musculoskeletal and connective tissue disorders: myalgia, muscle spasms, muscular weakness
Eye disorders: Frequency ‘not known’: Choroidal effusion, acute myopia, acute angle-closure glaucoma

SYMPTOMS AND TREATMENT OF OVERDOSAGE: Symptoms: There have been no reports of overdosage. Based on the pharmacological activities of APO-INDAPAMIDE (indapamide), overdosage may lead to excessive diuresis with electrolyte depletion. In cirrhotic patients, overdosage might precipitate hepatic coma.
Treatment: There is no specific antidote. Treatment is symptomatic and supportive. Discontinue drug. Induce emesis or perform gastric lavage. Correct dehydration, electrolyte imbalance, hepatic coma and hypotension by established procedures.

DOSAGE: One 2.5mg tablet per day taken in the morning as a single dose. If the response is not satisfactory after 4 to 8 weeks, increase in dosage is not recommended. Instead a non-diuretic antihypertensive agent should be added to the drug regimen. Alternatively, if in the opinion of the physician, an important diuretic effect is desirable for the patient's control, a different diuretic which allows for dose titration could be tried instead of indapamide.
In cases where two drugs are considered necessary to initiate treatment, the dose of APOINDAPAMIDE (indapamide) remains 2.5mg per day, taken in the morning as a single dose.

ADMINISTRATION ROUTE:
Oral

AVAILABILITY OF DOSAGE FORMS:
Apo-Indapamide 2.5mg: Each pink, round, biconvex, film-coated, engraved APO on one side, 2.5 on the other side, contains 2.5mg indapamide. Available in Aluminium/PVC blisters of 30 tablets.

STORAGE CONDITIONS:
Store below 30°C

Date of Revision of Package Insert: 3rd March 2026

Manufacturer: Apotex Inc, 150 Signet Drive, Weston (Toronto), Ontario, Canada, M9L 1T9.

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